MediFlow

A data engineering idea: from raw hospital files to trusted analytics

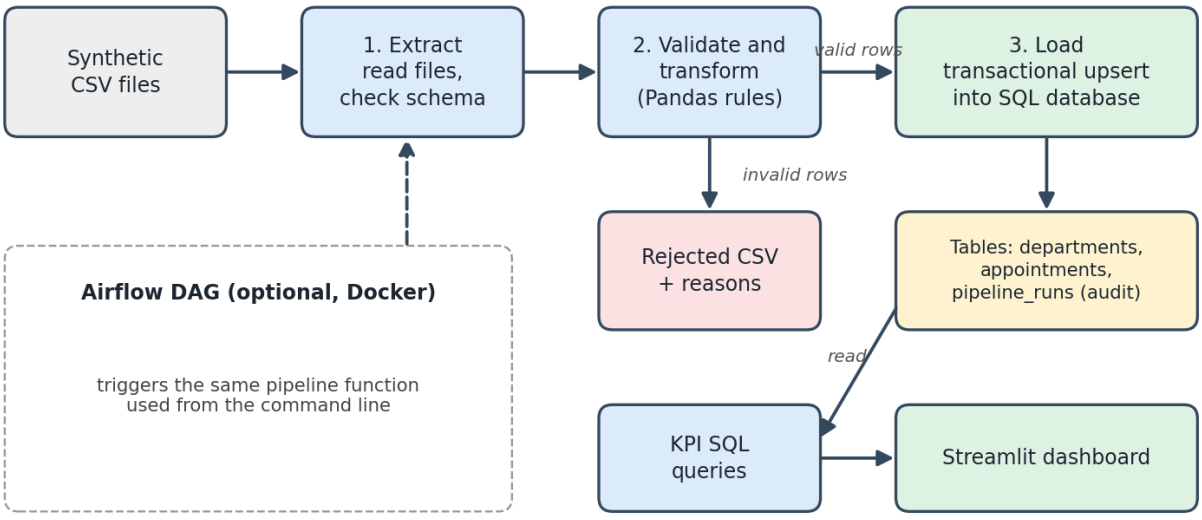
The problem. Hospitals export appointment data as flat files, and those files are rarely clean: missing values, duplicates, wrong statuses, impossible timestamps. A report built directly on them can show wrong numbers.

The idea. Put a reliable data pipeline between the raw files and the report. The pipeline extracts the data, checks every record, separates good rows from bad ones, stores the good rows in a SQL database, and calculates KPIs from that database.

> All healthcare records are synthetic and are intended exclusively for software development, education, and demonstration. This is an educational operations-analytics idea, not a clinical decision-support system.

Proposed pipeline

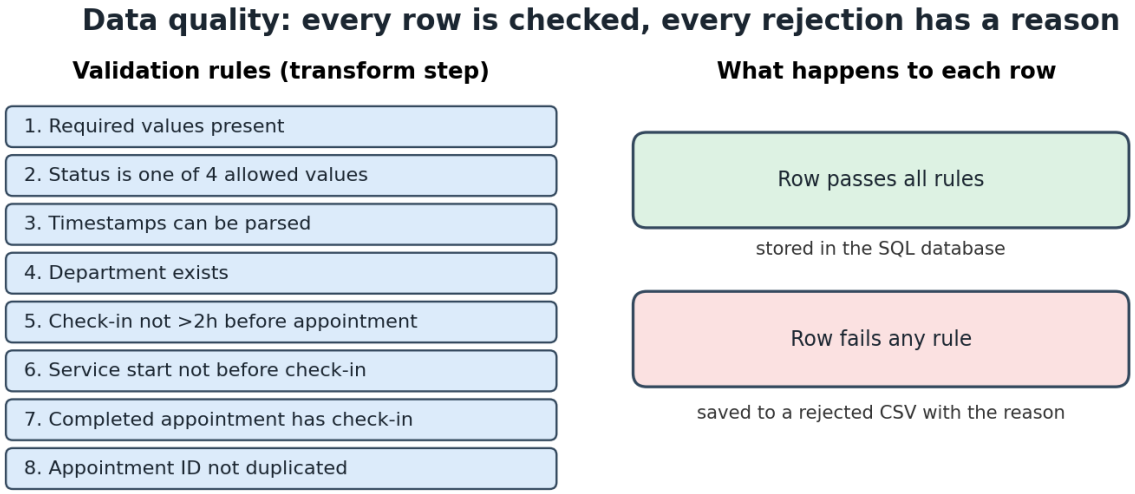
MediFlow pipeline: from raw CSV files to KPIs



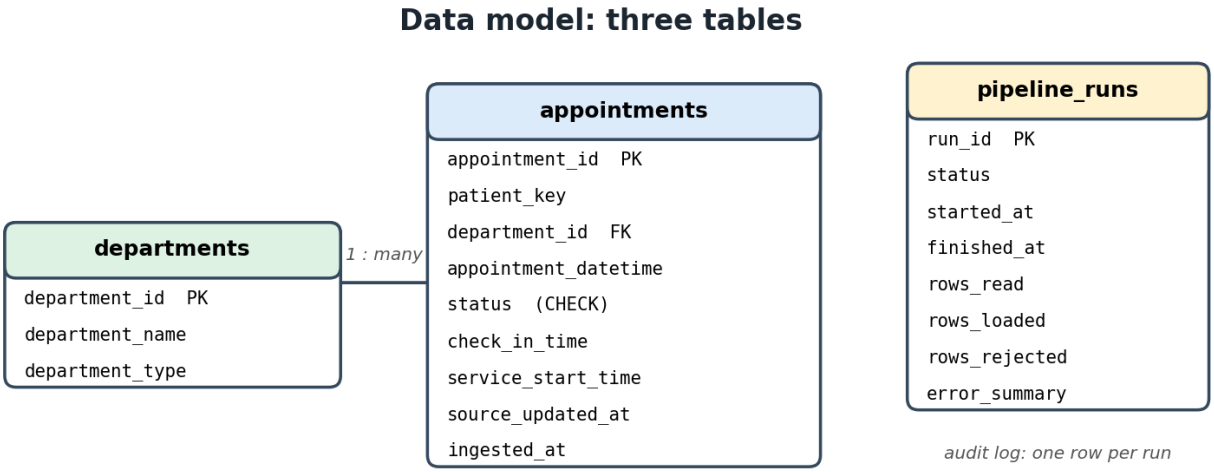
Step	Purpose
Extract	Read the files and confirm the expected columns exist
Validate and transform	Apply business rules; keep accepted and rejected rows apart
Load	Upsert accepted rows so re-running never duplicates
Analyze	Calculate KPIs with SQL
Monitor	Log every run and show results in a dashboard

Data quality by design

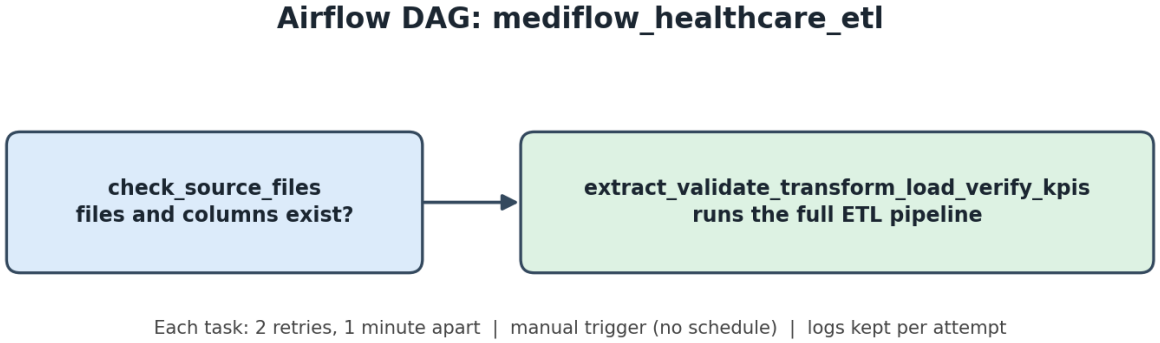
Rejected rows are never silently dropped: each one is kept together with the reason it failed.



Data model



Orchestration



Apache Airflow would run the pipeline as two dependent tasks, with retries and logs for each attempt.

Questions the platform answers

KPI	Definition
Total and completed appointments	Counts
No-show rate	No-Show / (Total - Cancelled)
Cancellation rate	Cancelled / Total
Average and median waiting time	Service start minus check-in, in minutes
Volume by department and by day	Grouped counts
Pipeline health	Rows read, loaded, rejected; success or failure

Proposed technologies

Technology	Role
Python and Pandas	Pipeline logic and validation
SQLite / PostgreSQL	Local and shared relational storage
SQLAlchemy	Database access
Apache Airflow and Docker Compose	Orchestration and reproducible runtime
Streamlit	Dashboard
pytest	Automated tests

Concepts demonstrated: ETL, data quality, data modeling, incremental loading, idempotency, orchestration, SQL analytics, automated testing.

DEPI Data Engineering capstone, for educational and portfolio purposes.